

Nuclear Not for a Small Island

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See last slide for references



Fianna Fail Think Nuclear will solve our energy Challenges !!



Does Nuclear work for a Small Country

- Government has to cover the insurance
 - So one Fukushima accident would bankrupt us.
 - 150 Billion Euro over 30 years to clean it up
- Eggs in one basket
 - Shutting down the plant is covered by interconnector and batteries
 - But a design recall or an accident means long term loss of all nuclear on the grid – and it is too big a percentage
 - France has 56 power stations with 4 different designs

Waste !?

- 500,000 metric tonnes of nuclear waste worldwide
- ... in temporary storage
- Waiting for someone else to solve it
- Or to dig a hole deep enough.

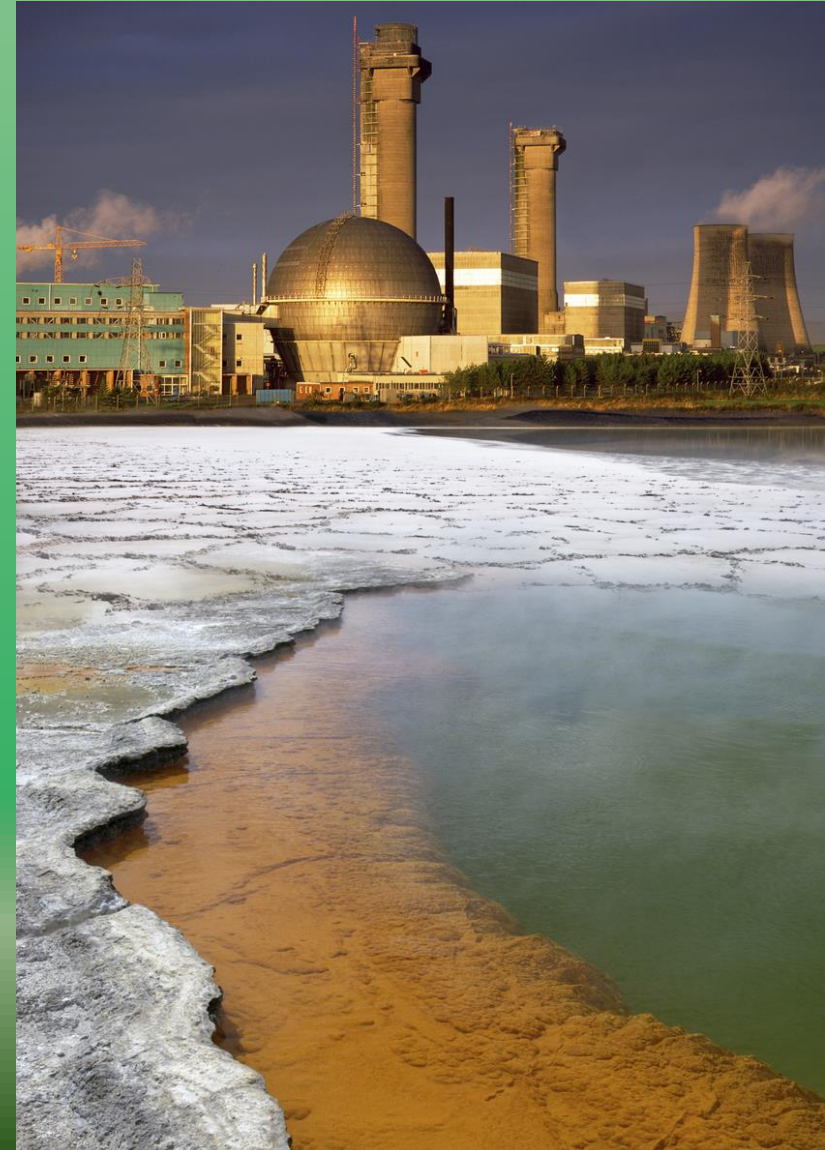
Waste Storage - Dry Casks

- Waste storage after 8 years in water
- Casks last 100 years
- Some are 40 years old!



Waste - the Windscale / Sellafield Debacle

- Fire in Windscale in 1957 was their low point.
- A leak of 2 cubic metres of water per day discovered in 1970's
- Still leaking today !
- It will be sorted by 2050 when the pools are drained



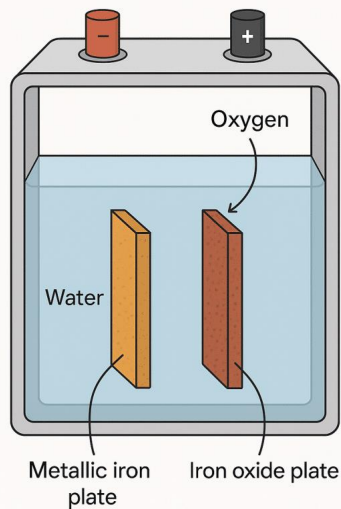
Reliability

Renewables

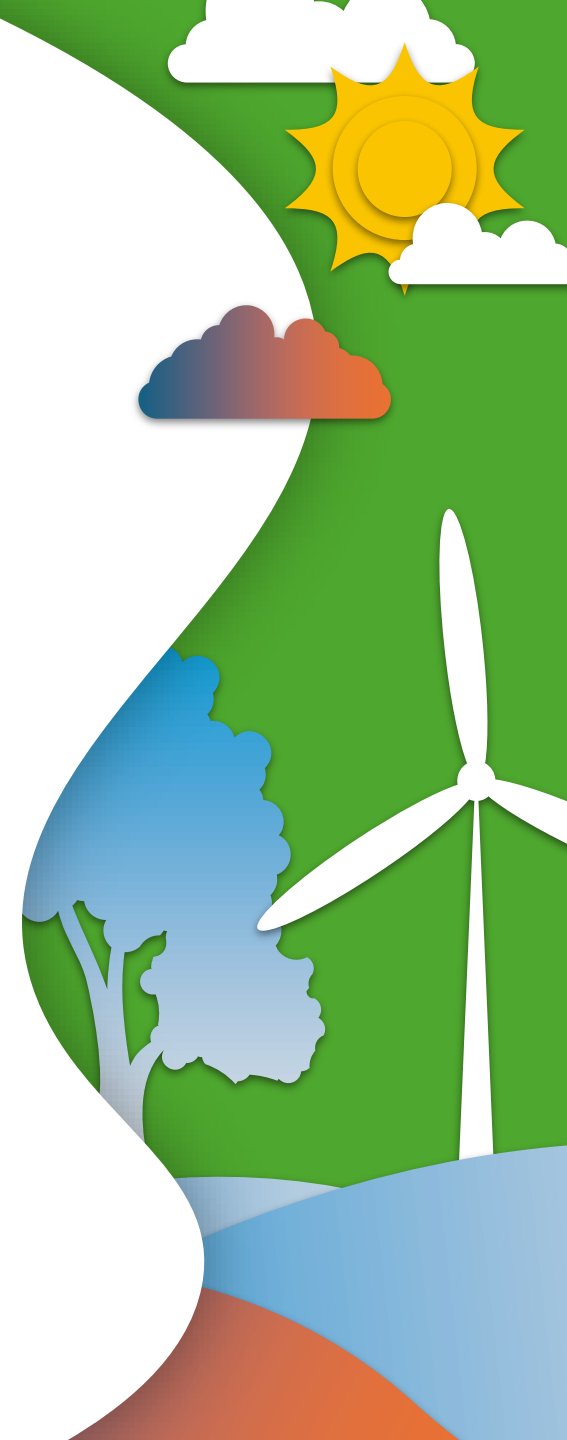
Sometimes the wind does not blow, and sun does not shine

Short Term

Iron-Air Battery



Batteries : alternatives to Lithium can give 100 hours backup



Reliability

Renewables

Sometimes the wind does not blow, and sun does not shine

Long Term - Seasonal

Interconnectors



Hydrogen or Ammonia storage



Reliability

Nuclear

Always on – no supply issues



Reliability

Nuclear

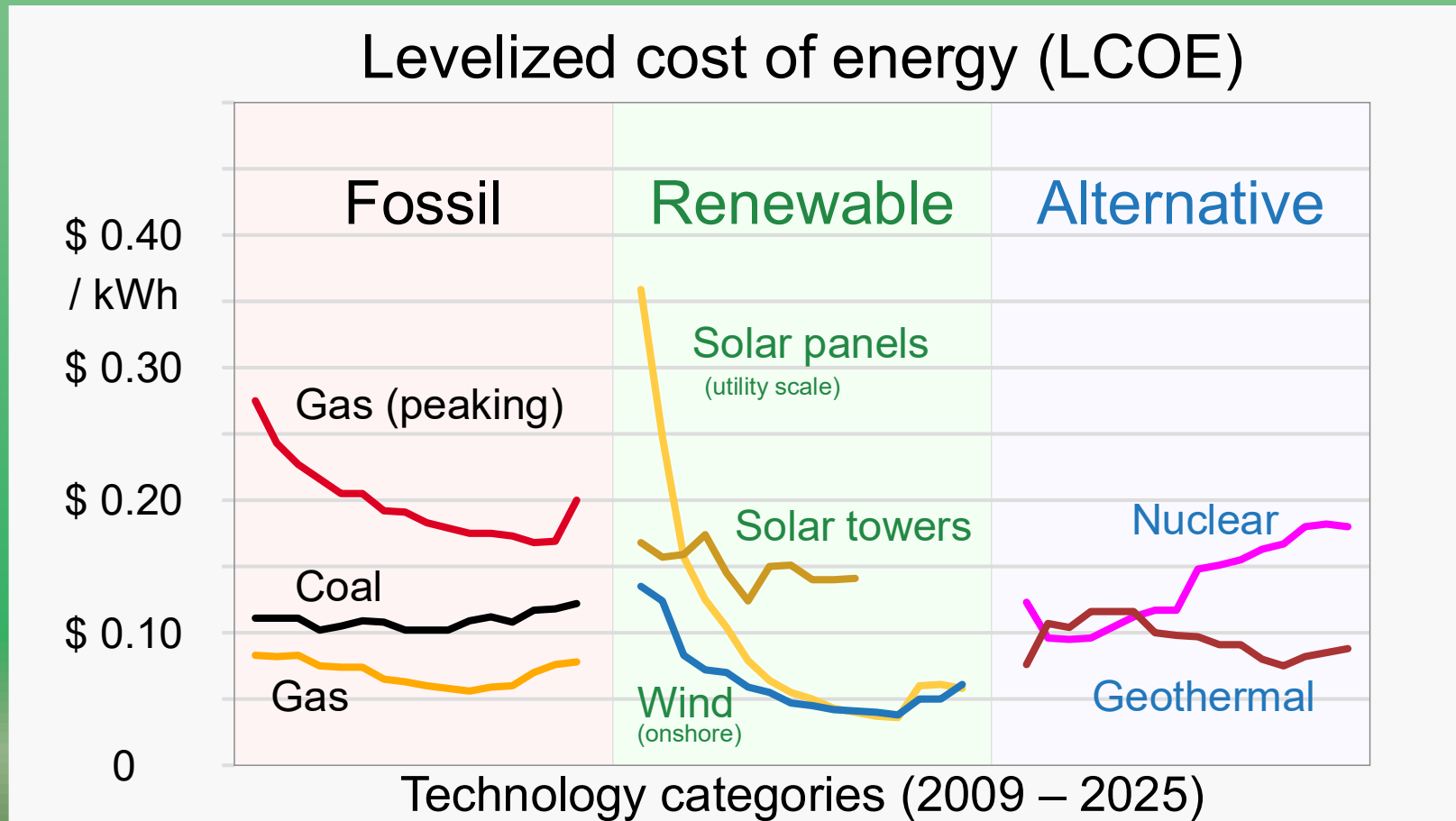
Always on – no supply issues

Except ...

The Siene and the Rhone rivers are so warm that they can not cool the nuclear stations and they have been shut down



It's the Economy Stupid !



A timeline for Ireland

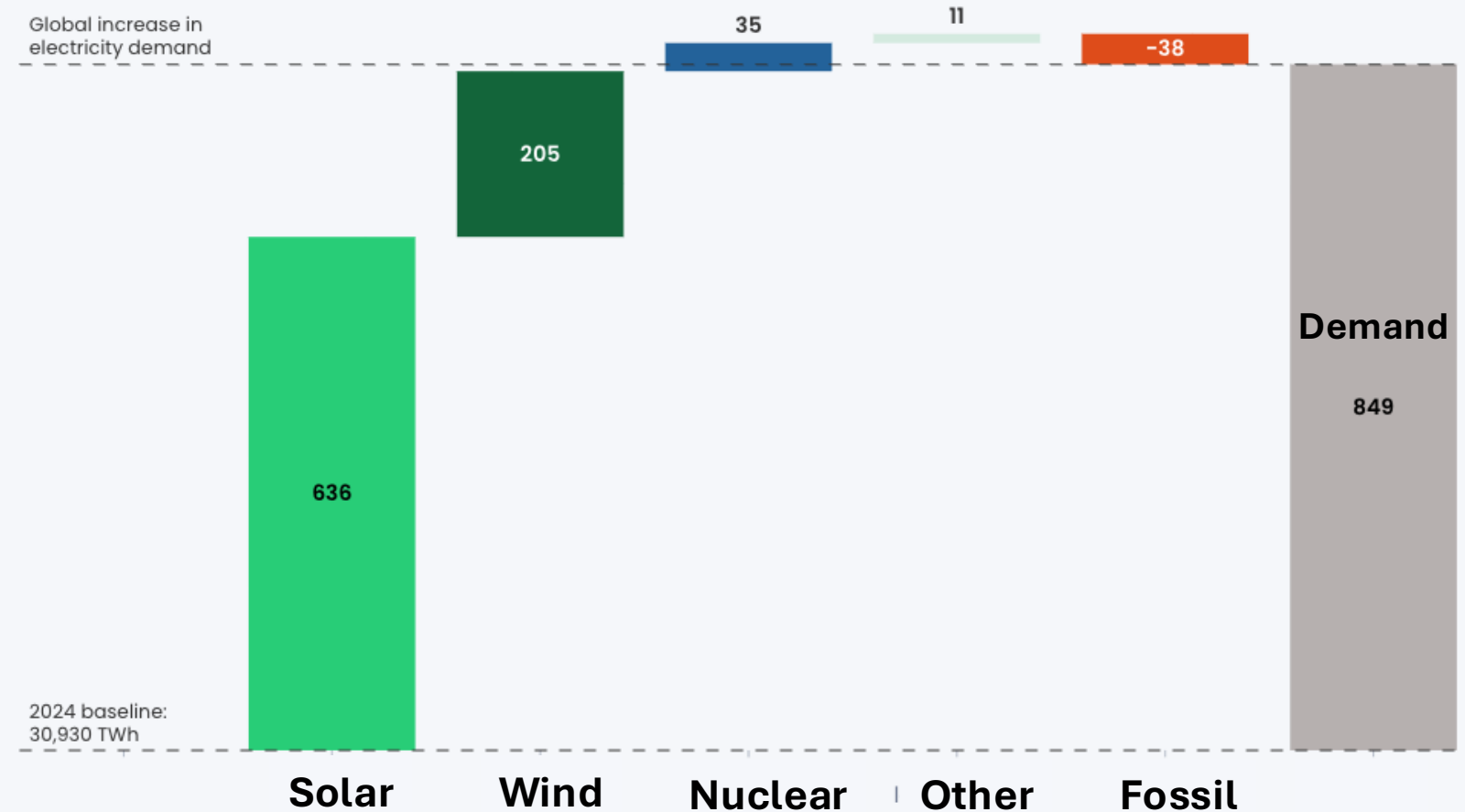
2021–2022 ~1 year	Citizens' Assembly
2022–2030 ~8 years	Wait for commercial SMRs
2030–2035 ~5 years	Political process, Dáil, referendum, legislation
2035–2040 ~5 years framework updates	Policy, environmental assessment, planning
2040–2047 ~7 years review	Planning permission, appeals and judicial
2047–2059 ~12 years financing	Licensing, regulator, grid connection,
2059–2066 ~7 years	Construction and commissioning
Total 45 years	2021 → 2066

The squeezing Out Problem



Do we need
nuclear because
renewables
cannot keep up
with demand -
NO

Change in electricity generation in 2025 vs. 2024 (TWh)



Source: Yearly electricity data, Ember

References

China installing enough solar to compete with nuclear plants

<https://www.theguardian.com/world/2025/jun/26/china-breaks-more-records-with-massive-build-up-of-wind-and-solar-power>

Hot Mess: The timeline to building a nuclear power plant in Ireland.

<https://www.rte.ie/radio/radio1/clips/22035642/>

Shutting down reactors due to hot weather <https://www.euronews.com/2026/06/25/france-takes-nuclear-reactors-offline-amid-record-heatwave>

Finna Fail and Nuclear vs. Solar Farms <https://shows.acast.com/irish-examiner-podcasts/episodes/does-ireland-need-nuclear-power-with-john-gibbons>

Compare cost of power based on source : https://en.wikipedia.org/wiki/Levelized_cost_of_electricity

Iron-Air Batteries in Donegal: <https://ballynahoneenergystorage.ie/>

Solar can keep up with demand increase:

<https://ember-energy.org/latest-insights/global-electricity-review-2026/2025-in-review/>